

Spring 2026 Practice Final Exam

Foundations of Data Science

Name

Instructions

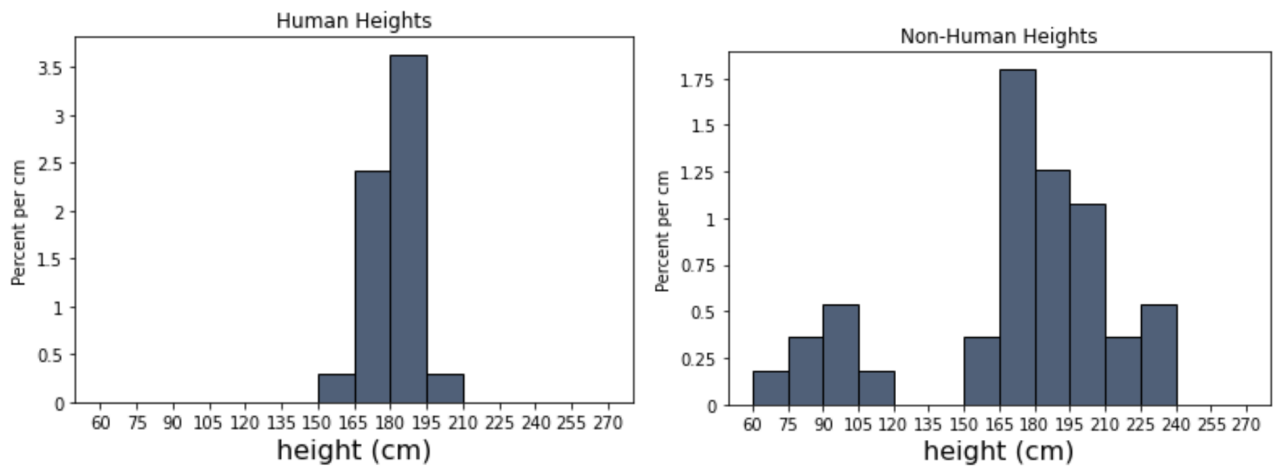
- Open the Final Exam Reference Guide (linked in Canvas). We will provide you with a new copy of this reference to use on the exam.
- Previews of most of the tables you will work with in this exam are at the end of exam in the Table Reference section.
- You can assume the following code has been run, when you are writing your Python code:

```
from datascience import *
import numpy as np
import matplotlib+
%matplotlib inline
import matplotlib.pyplot as plt
plt.style.use('fivethirtyeight')
```

- Select the correct response(s) or provide a written response depending on the question type. If a prompt implies you should write code, then you can provide your own code or use the provided template. Try to provide your responses in the template blanks or boxed spaces provided. If you find that you need additional space, write your extended response(s) on a blank sheet of paper and number them, so we can connect your response to the question.
- Once you have made an attempt on the exam, upload a PDF of your attempt to Canvas for a Complete score. Sample solutions will be released before the Final Exam Study Session.
- You will be asked a few questions about the Final Project (Project 2). Some of the questions may be about the code you wrote, and some of the questions may have you interpret the results and visualizations.

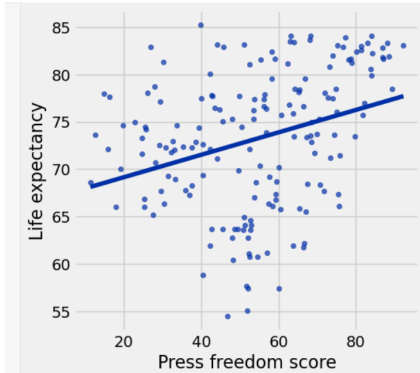
This practice exam contains questions from past UC Berkeley DATA 8 final exams in addition to a few additional or modified questions.

- The median income from one group of US residents (Group 1) is \$58,600 while the median income for another group of US residents (Group 2) is \$42,500. What can we say for sure based on these statistics? **Select all that apply.**
 - ☐ Being identified with Group 2 will cause you to earn less than if you were identified with Group 1.
 - ☒ **There is an association between being identified with the groups and median incomes in the US.**
 - ☐ There is a statistically significant difference between the median incomes of these two groups.
- What is a purpose of Bayes' Rule? **Select all that apply.**
 - ☒ **To quantify the impact of subjective probabilities on our predictions.**
 - ☐ To test whether there is a causal relationship between two variables.
 - ☐ To evaluate the accuracy of a machine learning model on the population.
 - ☐ To determine what percentage of our data lies within a certain number of standard deviations from the mean.
 - ☒ **To update our predictions with new information.**
- The following histograms display the distribution of the heights of humans and non-humans based on the Star Wars character data found in the table `characters`. Respond to the following prompts based on these visuals.



- Based on the plot shown for height of humans above, the mean human height is similar to the median human height. **Select one.**
 - ☒ **True**
 - ☐ False
 - ☐ This is not possible to determine from the graphic.
- According to this visualization, the standard deviation of non-human heights is larger than the standard deviation of human heights. **Select one.**
 - ☒ **True**
 - ☐ False
 - ☐ This is not possible to determine from the graphic.

4. **Global data.** The following scatterplot shows the relationship between a country's "Press freedom score", with a higher score indicating more press freedom, and its "Life expectancy", in years. See the table reference for a preview of the table `press_life_table`. The scatterplot also shows the linear regression model.



Not all countries have data in this table, so we will use this sample to conduct a hypothesis test for the slope of the linear model for the population of all countries. The following are our hypotheses.

Null hypothesis: The slope of the linear regression for the population is 0.

Alternative hypothesis: The slope of the linear regression for the population is not 0.

- (a) We bootstrap our sample 5000 times to get the following 95% confidence interval for the slope: (0.074, 0.164). Which of the following statements can we conclude? **Select all that apply**
- ☒ **95% of the slopes of the bootstrapped samples are between 0.074 and 0.164.**
 - ☐ There is a 95% chance that the slope of the linear regression for the population is between 0.074 and 0.164.
 - ☒ **We have evidence to reject the null hypothesis, and support the claim that the slope of the linear regression for the population is not 0.**
 - ☐ We do not have evidence to reject the null hypothesis that the slope of the linear regression for the population is 0.
- (b) The RMSE for the linear model is approximately 6.91. You decide to fit a quadratic model $ax^2 + bx + c$ to the data. You create the following function to calculate the quadratic RMSE:

```
def quad_rmse(a, b, c):  
    x = press_life_table.column('Press freedom score')  
    y = press_life_table.column('Life expectancy')  
    quad_estimates = a * x ** 2 + b * x + c  
    quad_residuals = y - quad_estimates  
    return np.sqrt(np.mean(quad_residuals ** 2))
```

Write code to find the best fitting quadratic model for the `press_life_table` data. Specifically, the code should produce the values for `a`, `b`, and `c`.

Sample Solution:

```
a, b, c = minimize(quad_rmse)  
a, b, c
```

- (c) It turns out that the quadratic model fits the data a little bit worse than the linear model, in terms of RMSE. What does this mean? **Choose one.**
- ☐ The RMSE for the quadratic model is less than 6.91.
 - ☐ The RMSE for the quadratic model is equal to 6.91.
 - ☒ **The RMSE for the quadratic model is more than 6.91.**

5. **Movies.** Rotten Tomatoes, a movie review website, is measuring which of the two movies - Hoppers or The Super Mario Galaxy Movie - has higher reviews among Berkeley students. They believe that Berkeley students will give higher reviews to the Hoppers movie.

- (a) Researchers at Rotten Tomatoes randomly sample 1000 Berkeley students and show each student both movies under identical viewing conditions. Immediately after watching each movie, every student is asked to rate that movie on an integer scale from 1 (worst) up to, and including 10 (best). The reviews are collected in a table named `reviews`; previewed in the Table Reference section. This study design is an experiment. **Choose one.**

☒ **True** ☐ False

- (b) Which of the following is a correct null hypothesis that Rotten Tomatoes should use to assess their claim? **Select one.**

- ☐ The Hoppers movie has a different distribution of reviews than the Super Mario Galaxy movie among the given sample of Berkeley students.
- ☐ The Hoppers movie has the same distribution of reviews as the Super Mario Galaxy movie among the given sample of Berkeley students.
- ☐ The Hoppers movie has a different distribution of reviews than the Super Mario Galaxy movie among Berkeley students.

☒ **The Hoppers movie has the same distribution of reviews as the Super Mario Galaxy movie among Berkeley students.**

- (c) Please state a clear and complete alternative hypothesis that Rotten Tomatoes should use to assess their claim.

Sample Solution: The Hoppers movie has higher reviews than the Super Mario Galaxy movie among Berkeley students.

- (d) Using the table `reviews` (see Table Reference), provide code that will generate a histogram showing the distribution of `'review'` values for Hoppers overlaid with a histogram of `'review'` values for the Super Mario Galaxy movie.

Sample Solution:

```
reviews.hist('review', group='movie')
```

- (e) If there is NO association between the movie and the review values, then the histograms will almost perfectly overlay each other. **Choose one.**

☒ **True** ☐ False

- (f) Rotten Tomatoes uses the **difference of means** as their test statistic. Complete the function below so that it returns the difference of mean reviews between the two movies. Larger values of the test statistic should favor the alternative hypothesis. **Note:** Assume that the `reviews_table` argument resembles the `reviews` table in the Table Reference.

Hint: The `group` function will return a table that is sorted alphabetically based on the values in the column used for grouping.

```
def test_statistic(reviews_table):  
    means_col = _____  
    return _____
```

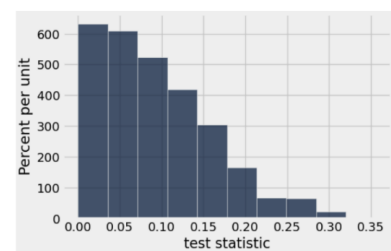
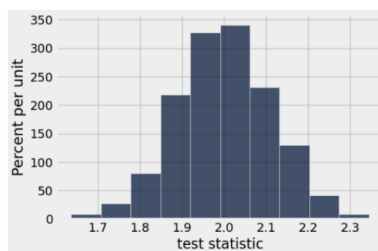
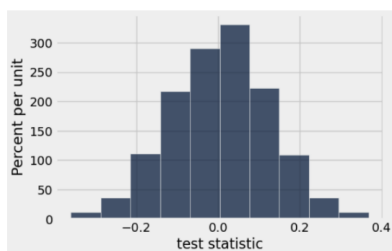
Sample Solution:

```
def test_statistic(reviews_table):  
    means_col = reviews_table.group(0, np.mean).column(1)  
    return abs(means_col.item(1) - means_col.item(0))
```

- (g) Which of the following may be used to create simulations under the null hypothesis? **Select all that apply.**

- ☒ **Shuffle the values of only the 'movie' column.**
- ☒ **Shuffle the values of only the 'review' column.**
- ☒ **Shuffle the values of the 'movie' column, then shuffle the values of the 'review' column.**
- ☐ Randomly sample all of the rows of the reviews table **with replacement**.
- ☐ Randomly sample all of the rows of the reviews table **without replacement**.
- ☐ None of the above.

- (h) Suppose we simulate 10,000 values of the test statistic under the null hypothesis. Which of the following will our distribution of simulated test statistics most closely resemble? **Select one.**



- ☒ **The left graph**
- ☐ The middle graph
- ☐ The right graph

- (i) Suppose to produce the distribution of 10,000 simulated test statistics you chose above, the simulated differences in means were stored in an array called `simulated_diffs`. If the observed difference is named `observed_diff`, write code that will calculate the p -value for this hypothesis test using the simulated data.

Sample Solution:

```
np.count_nonzero(simulated_diffs >= observed_diff) / 10_000
```

- (j) You obtain a p -value of 0.37 from your experiment above. Which of the following statements are true? **Select all that apply.**

Note: Recall that larger values of your test statistic should favor the alternative hypothesis.

- ☒ **Your observed test statistic lies at the 63rd percentile of the distribution of test statistics simulated under the null hypothesis.**
- ☐ 37% of the test statistics simulated under the null hypothesis were as, or less extreme than the observed test statistic.
- ☐ The Super Mario Galaxy movie has higher reviews than the Hoppers movie among Berkeley students.
- ☒ **With a p -value cutoff of 5%, our data are consistent with the null hypothesis.**
- ☐ None of the above.

- (k) Which of the following statements are true? **Select all that apply.**

- ☒ **If Rotten Tomatoes repeats the same experiment, but instead, they sample 10,000 Berkeley students, the observed test statistic will more accurately reflect whether Hoppers is reviewed higher than Super Mario Galaxy among Berkeley students.**
- ☒ **If Rotten Tomatoes repeats the same experiment, but instead, they sample 10,000 Berkeley students, the distribution of test statistics simulated under the null hypothesis will have a smaller standard deviation.**
- ☐ If Rotten Tomatoes repeats the same experiment, but instead, **they simulate 1000 values of the test statistic under the null hypothesis**, the distribution of these simulated test statistics will have a larger standard deviation.
- ☐ None of the above.

6. **Spotify.** Christina is interested in learning more about the duration of songs on Spotify. She collects a random sample of 400 songs listed on the platform and stores the data in the table `songs`, which has one column labelled “Duration”. The average song duration in the sample is 185 seconds and the standard deviation is 25 seconds. Christina wants to use this sample of songs to make some estimates about the population of songs and their durations.

- (a) Define a function `song_ci` that constructs a 95% confidence interval for the population mean as follows and returns it as an array. The function takes in the argument `reps`, the number of bootstrap repetitions wanted.

```
def song_ci(reps):
    stats = _____
    for _____:
        resample = _____
        new_mean = _____
        stats = _____
    left_end = _____
    right_end = _____
    return _____
```

Sample Solution:

```
def song_ci(reps):
    stats = make_array()
    for i in np.arange(reps):
        resample = songs.sample()
        new_mean = np.mean(resample.column(0))
        stats = stats.append(stats, new_mean)
    left_end = percentile(2.5, stats)
    right_end = percentile(97.5, stats)
    return make_array(left_end, right_end)
```

- (b) Which of the following conclusions can you draw using the 95% confidence interval generated in the previous part a? **Select all that apply.**

- ☐ The duration of 95% of songs in the population are in the interval generated.
- ☐ There is a 95% chance that the population’s true mean duration is within the interval generated.
- ☐ There is a 95% chance that the sample’s true mean duration is within the interval generated.
- ✓ **We are 95% confident that the population’s true mean duration is within the interval generated.**
- ✓ **There is a 95% chance that the next 95% confidence interval we generate will contain the population’s true mean duration.**

(c) Christina creates an interval by using `song_ci(10000)`. To get a more accurate estimate at the same level of confidence, Christina would like to create a new 95% confidence interval that is half as wide as this one. Which one of the following do you think is the best advice for her?

- ☐ She should use `song_ci(20000)`.
- ☐ She should use a sample of size 800.
- ☐ She should use `song_ci(40000)`.
- ✓ **She should use a sample of size 1600.**

7. **Stocks.** Waystar is an organization that owns businesses and trades stocks in a variety of sectors including media, entertainment, tech, etc. Its executive team wants to understand the performance of its businesses. Waystar executives Siobhan and Roman put together a table called `performance` (previewed in the Table Reference section), which contains randomly sampled public information about the various businesses' performance over the last 40 years. Two of the columns are 'Profit' (float) and 'Year' (int).

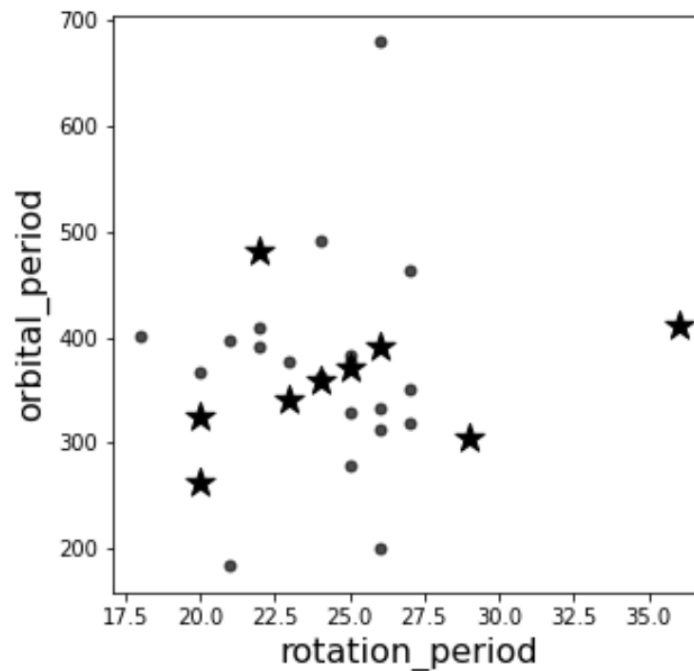
To forecast Waystar's performance, Siobhan wants to understand what the profits for the businesses might be in future years. She first creates a function called `correlation`, which returns the correlation between two numerical arrays. Complete her `predict_profit` function, which takes in a `year_of_interest` (int), `years` (np.array), `profits` (np.array), and returns the predicted profit for that provided year of interest. Note that the arrays `years` and `profits` can be assumed to have come from a table like `performance` (same columns and data types).

```
def predict_profit(year_of_interest, years, profits):
    r = correlation(_____)
    slope = r * np.std(profits) / np.std(years)
    intercept = _____
    predicted_profit = _____
    return predicted_profit
```

Sample Solution:

```
def predict_profit(year_of_interest, years, profits):
    r = correlation(years, profits)
    slope = r * np.std(profits) / np.std(years)
    intercept = np.mean(profits) - slope * np.mean(years)
    predicted_profit = intercept + slope * year_of_interest
    return predicted_profit
```


8. **Star Wars.** Use a k-NN classifier to predict whether or not a planet in the Star Wars universe contains a temperate climate. The following scatter plot indicates data points associated with planets that contain a temperate climate with a solid dot and data points associated with planets without a temperate climate with a star. Each planet's rotation and orbital periods were used to plot the dots and stars.



- (a) The planet Dorin has a rotation period of 22 and an orbital period of 409. The planet Endor has a rotation period of 18 and an orbital period of 402. Based on these values, write an arithmetic expression that Python can evaluate to calculate the distance between these two data points. (Optionally, provide a short explanation to help us consider partial credit in scoring.)

Sample Solution: `np.sqrt((18 - 22) ** 2 + (402 - 409) ** 2)`

- (b) What `has_temperate_climate` label would a k-NN classifier with `k = 5` assign to a planet with a rotation period of 22.5 and an orbital period of 600? (True corresponds with a dot, False corresponds with a star)
- ☒ True
- ☐ False

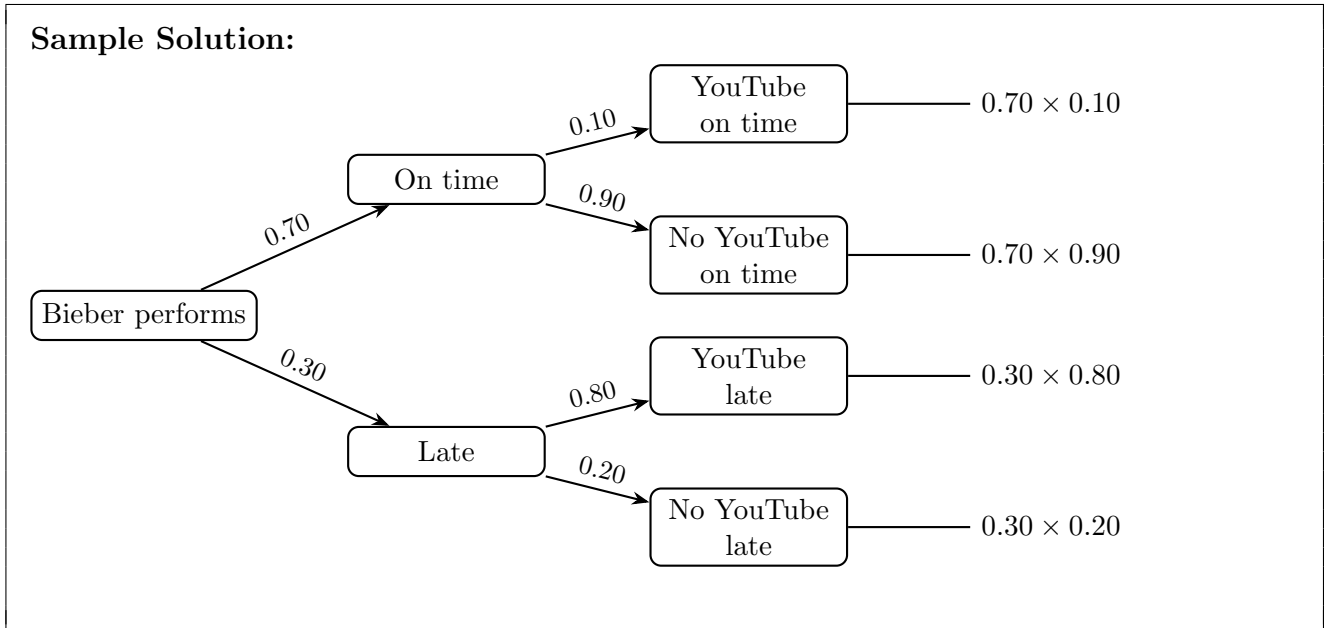
- (c) The tables `training_planets` and `testing_planets` are randomly created from all the available planets that have a column labeled as `temperate` with values `True` or `False`. The data in the `training_planets` table is visualized in the previous scatter plot. All of the test data is shown in the `testing_planets` table. (See Table Reference)

The k-NN classifier with $k = 5$ predicated a `True` label (predicting that they would have a temperate climate on at least part of the planet) for all the planets in the provided test data. What would be the accuracy of the classifier in this case? Express your answer as a fraction or decimal that Python can evaluate. (Optionally, provide a short explanation to help us consider partial credit in scoring.)

Sample Solution: 6 / 10

9. **Real Estate.** A real estate company has a dataset of all their buildings, with three attributes for each building: its size (in square feet), its type (residential or commercial), and its estimated value (sale price) if sold (in dollars). The standard visualization to understand the distribution of building types is: **Choose one.**
- ☒ **A bar chart** ☐ A line plot ☐ A scatter plot ☐ A histogram
10. **True or False.** Select True or False for each of the following:
- (a) The height of each bar in a density histogram represents the proportion of data within the corresponding bin. **Choose one.**
☐ True ☒ **False**
- (b) For any distribution, the percentage of data that lies beyond two standard deviations on either side of the mean is at most 2.5%. **Choose one.**
☐ True ☒ **False**
- (c) A classifier is considered to be overfitting if it performs very well on the test set. **Choose one.**
☐ True ☒ **False**
- (d) If we use linear regression to predict y-values based on our x-values, the mean of our residuals is zero. **Choose one.**
☒ **True** ☐ False
- (e) For any two events A and B , the probability $P(A \text{ and } B)$ is less than or equal to the probability $P(A \text{ or } B)$ **Choose one.**
☒ **True** ☐ False

11. **Coachella.** In the year 2031, Amy and Ron are excited to see Justin Bieber perform at Coachella for the 6th year in a row. Based on historical data, they know there's a 70% chance he will be on time, and otherwise he will be late. If he is on time there is only a 10% chance he will play music from YouTube, but if he is not on time then there is a 80% chance he will play music from YouTube.
- (a) Draw a tree diagram for this situation relating Justin Bieber's arrival possibilities and whether he plays music from YouTube or not.



- (b) What is the probability that Justin is on time and plays music from YouTube?
You can leave your answer as a mathematical or python expression.

Sample Solution: 0.70×0.10

- (c) What is the probability that he does not play music from YouTube? **You can leave your answer as a mathematical or python expression.**

Sample Solution: $0.70 \times 0.90 + 0.30 \times 0.20$

- (d) Suppose you can see the future and know he will play music from YouTube. Which of the following represents the probability that Bieber was on time, given that he played music from YouTube?

Sample Solution: $\frac{0.70 \times 0.10}{0.70 \times 0.10 + 0.30 \times 0.80}$

12. **Fitness.** Farhana likes to attend a popular class at the local fitness center. She collects data about each class she attends in a table called **workouts**. The table is previewed in the Table Reference section.

(a) Choose the best technique to answer the question "How high will Farhana's exercise heart rate be today given that 30 people attended class?" **Choose one.**

- ☐ Classification
- ☒ **Linear Regression**
- ☐ Hypothesis Test
- ☐ Randomized Control Experiment
- ☐ Bayes' Rule

(b) Choose the best technique to answer the question "Is there a difference in Farhana's heart rate between sunny and rainy days?" **Choose one.**

- ☐ Classification
- ☐ Linear Regression
- ☒ **Hypothesis Test**
- ☐ Randomized Control Experiment
- ☐ Bayes' Rule

(c) Choose the best technique to answer the question "What are the most likely weather conditions given that 12 people attended class today?" **Choose one.**

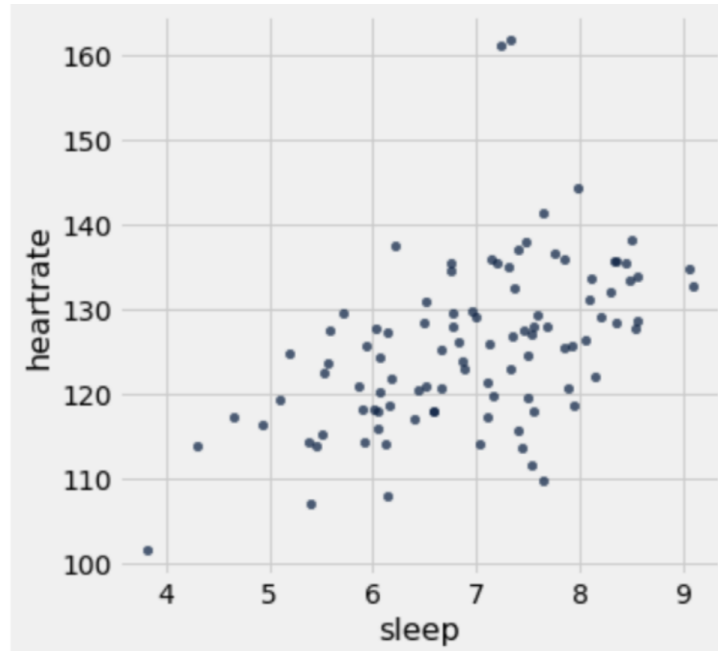
- ☒ **Classification**
- ☐ Linear Regression
- ☐ Hypothesis Test
- ☐ Randomized Control Experiment
- ☐ Bayes' Rule

(d) Choose the best test statistic for the following alternative hypothesis. **Choose one.**

Alternative Hypothesis: "The class size is larger on sunny days than it is on rainy days."

- ☐ The total variation distance between the class size distribution of sunny days and class size distribution of rainy days
- ☐ The empirical mean of class size on sunny days
- ☐ The empirical mean of class size
- ☒ **The difference of mean class size between sunny and rainy days**
- ☐ The difference of mean class size between sunny and cloudy days

- (e) Farhana wants to see if there is a relationship between how much sleep she gets and her heart rate during class, so she creates the following scatter plot.



Write a line of code that would generate the scatter plot above.

Sample Solution:

```
workouts.select('sleep', 'heartrate').scatter('sleep')
```

- (f) Which of the following are valid conclusions from this graph? **Choose all that apply.**
- ☒ **There is a positive association between her sleep and her heart rate during class**
 - ☐ There is a negative association between her sleep and her heart rate during class
 - ☐ Getting more sleep causes Farhana to have a higher heart rate during class
 - ☐ Getting less sleep causes Farhana to have a higher heart rate during class
 - ☐ Fewer people attend class on rainy days
- (g) Farhana asks her friend to compute the correlation coefficient r between these two variables, but her friend's code has at least one mistake in it. In the code below, circle and cross out each mistake and, if applicable, write the correct code immediately above. Alternatively, you can re-write the code in the solution box. You can use the following `standard_units` function from lecture:

```
def standard_units(any_numbers):  
    '''Convert any array of numbers to standard units.'''  
    return (any_numbers - np.mean(any_numbers))/np.std(any_numbers)
```

Here is her friend's code:

```
heartrate_in_su = standard_units('heartrate')
sleep_in_su = standard_units('sleep')
r = np.sum(heartrate_in_su * sleep_in_su)
```

Sample Solution:

```
heartrate_in_su = standard_units(workouts.column('heartrate'))
sleep_in_su = standard_units(workouts.column('sleep'))
r = np.mean(heartrate_in_su * sleep_in_su)
```

(h) Suppose we know the following:

- Farhana's heart rate has an average of 125 bpm and a standard deviation of 6 bpm
- Farhana's sleep has an average of 7 hours and a standard deviation of 1 hour
- The correlation between Farhana's heart rate and sleep is 0.5

If we were to fit a regression line to the scatterplot in (e), what would the predicted heartrate be when Farhana gets 8 hours of sleep? You may leave your answer as a mathematical/Python expression.

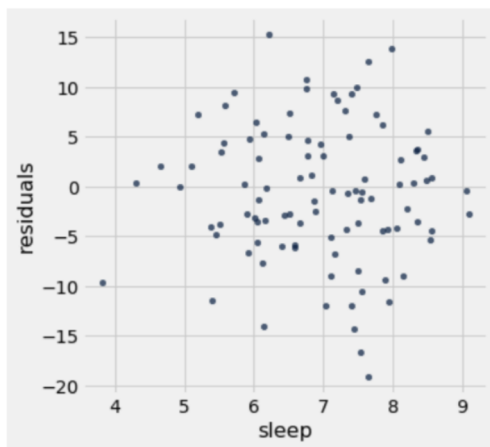
Sample Solution:

```
(0.5 * 6 / 1) * 8 + (125 - (0.5 * 6 / 1) * 7)
128 bpm
```

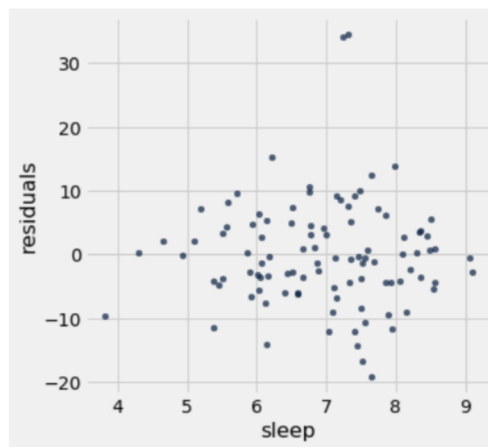
(i) Farhana begins a research apprenticeship in the School of Public Health, and wants to understand whether the amount of sleep someone gets causes a change in average heart rate during exercise. Her lab starts a study with a random sample of Berkeley undergraduate and graduate students who exercise regularly. Which of the following experiments would be able to answer her causal question? **Choose one.**

- ☐ Ask the undergraduates to sleep 7 hours per night and the graduate students to sleep 9 hours per night. Then, collect heart rate data during exercise.
- ☐ Randomly assign the subjects to two groups. Have the first group exercise for 1 hour per day, and have the second (control) group not exercise at all. Then, measure how much sleep they get before and after each exercise session.
- ☒ **Randomly assign the subjects to two groups. Have the first group sleep for 7 hours or less per night, and the second group sleep 9 hours or more per night. Then, collect heart rate data during exercise.**
- ☐ It is impossible to determine a causal link between these two variables.

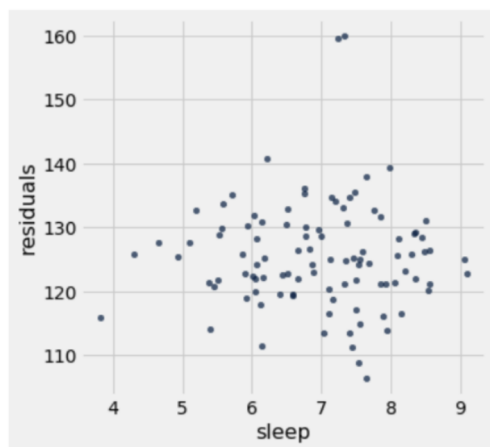
(j) Which of the following is the residual plot for the scatter plot shown in part (e)?



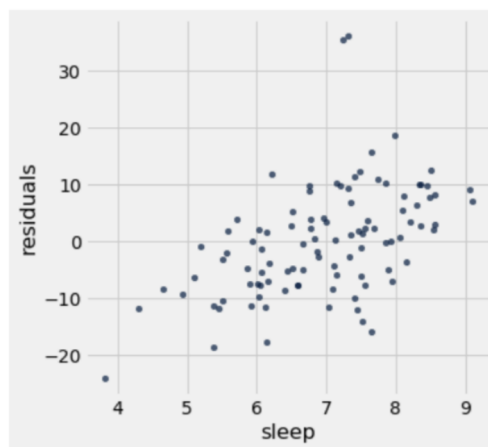
A



B

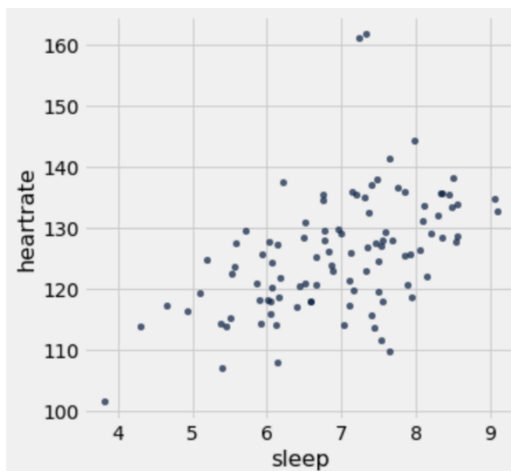


C



D

Here is the scatterplot for your convenience:



Choose one.

☐ Option A ☒ Option B ☐ Option C ☐ Option D

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Table Reference

characters

Here is a preview of the table characters:

name	height	mass	homeworld	human
Luke Skywalker	172	77	Tatooine	True
C-3P0	167	75	Tatooine	False
R2-D2	96	32	Naboo	False

... (45 rows omitted)

- **name:** (str) the name of the character
- **height:** (float) the height (cm) of the character
- **mass:** (float) the mass (kg) of the character
- **homeworld:** (str) the homeworld of the character
- **human:** (bool) an indicator if the character is human or not

global data

Here is a preview of the table press_life_table:

Country	Life expectancy	Press freedom score
Afghanistan	66.03	17.88
Albania	79.6	58.18
Algeria	76.26	44.64
Andorra	84.04	68.3
Angola	64.62	52.67

... (168 rows omitted)

reviews

Here is a preview of the table reviews:

movie	review
Super Mario Galaxy	6
Super Mario Galaxy	6
Hoppers	10
Hoppers	6
Hoppers	8

... (1995 rows omitted)

performance

Here is a preview of the table `performance`:

Name	Year	Revenue	Profit	Sector	Advice
Brightstar Cruises	2019	52.3	18.9	Entertainment	Hold
Adventure Parks	2018	42.8	16.3	Entertainment	Sell
Vaulter	2021	150.9	80.2	Tech	Buy

... (160 rows omitted)

- **Name:** (str) the name of the business
- **Year:** (int) the year of the financial performance
- **Revenue:** (float) the business's revenue (in millions of USD)
- **Profit:** (float) the business's profit margin (a percentage between 0 and 100)
- **Sector:** (str) the business's sector in the industry
- **Advice:** (str) Wall Street's stock purchase advice ('Buy', 'Hold', 'Sell')

testing_planets

Here is a preview of the table `testing_planets`:

name	rotation_period	orbital_period	has_temperate_climate
Polis Massa	24	590	True
Hoth	23	549	False
Ryloth	30	305	True
Glee Anselm	33	206	True
Kashyyyk	26	381	False
Tatooine	23	304	False
Felucia	34	231	False
Muunilinst	28	412	True
Geonosis	30	256	True
Cerea	27	386	True

workouts

Here is a preview of the `workouts` table and some information about the data values:

size	heartrate	weather	sleep
33	145.1	sunny	7.3
28	100.7	sunny	6.5
23	124	sunny	5
10	137.8	rainy	9

The table contains four columns:

- **size:** an int, the number of people who attended the class
- **heartrate:** a float, her average heart rate during the class in beats per minute (bpm)
- **weather:** a string, the weather conditions for that day
- **sleep:** a float, the number of hours of sleep she got the night before